

SYLLABUS :-

Basic Thermodynamics, ME2XXXX (New), 3-1-0-4 Introduction and fundamental concepts, system and surroundings, concept of control mass and control volume with examples, unit system, Temperature and zeroth law of thermodynamics, thermodynamic state, thermodynamic processes and thermodynamic equilibrium, Simple compressible substance, pure substances, phase and phase rule, Concept of energy, energy stored and energy in transit, definition of thermodynamic work and heat transfer, different modes of work transfer, First law of thermodynamics; cyclic and non-cyclic processes, concept of internal energy, first law of thermodynamics applied to an open system, concept of enthalpy, steady state steady flow energy equation, unsteady state energy equation, first law applied to simple components such as nozzles, diffusers, pumps and compressors, Second law of thermodynamics, the directional aspects of spontaneous processes, second law applied to cyclic and non cyclic processes, Clausius inequality, concept of entropy, entropy balance equations, concept of entropy increase, reversible and irreversible processes, Formal statements of second law, Concept of heat engines, refrigerators and heat pumps, Carnot cycle and absolute temperature scale, Carnot theorems, Kelvin-Planck and Clausius statements, Concept of availability and irreversibility, Thermodynamic properties, property relations, Maxwell relations, Clausius-Clayperon equation, specific heats and specific heat ratio, Joule Thomson coefficient and inversion temperature, Thermodynamic properties of pure substances and mixtures, equations of state, ideal and real gases and gaseous mixtures, Estimation of thermodynamic properties, sources of properties, tables, charts and property equations, use of various software (EES, RefProp etc.) for property evaluation, psychrometry, psychrometric properties and psychrometric chart, - 277 -Page 4 of 7

Basic thermodynamic cycles: Performance parameters, thermal efficiency, COP and second law efficiency, Simple vapour cycles for power generation and refrigeration – Carnot, Simple Rankine and simple vapour compression refrigeration (Evans-Perkins) cycles, Simple gas cycles for power generation and refrigeration – Carnot, Brayton, Joule, Stirling, Otto and Diesel cycles

Suggested Books: 1. Fundamentals of Thermodynamics, Richard Edwin Sonntag, Claus Borgnakke, Gordon John Van Wylen, Wiley, 2003. 2. Fundamentals of Engineering Thermodynamics, Michael J. Moran, Howard N. Shapiro, Daisie D. Boettner, Margaret B. Bailey, Wiley, 9th edition, 2018